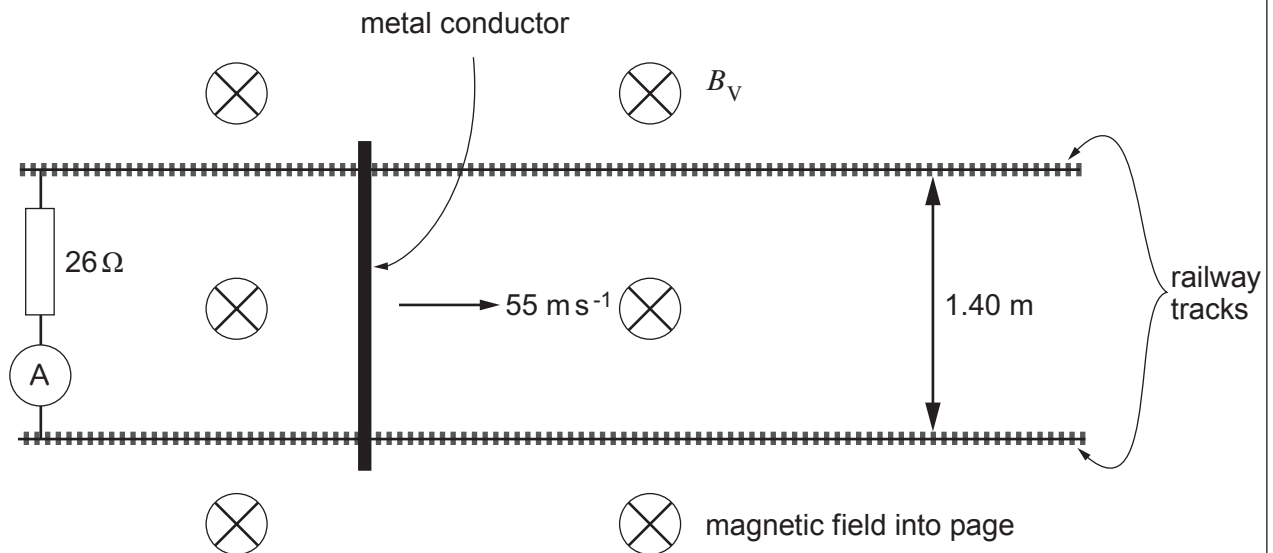


5. A metal conductor is placed across horizontal railway tracks and moved quickly in the direction shown. This set up is designed to measure the vertical component (B_V) of the Earth's magnetic field.



- (a) (i) Explain why a current is detected by the ammeter.

[2]

.....

.....

.....

.....

- (ii) State why the current is independent of the horizontal component of the Earth's magnetic field.

[1]

.....

.....

.....

- (b) Calculate a value for the vertical component of the Earth's magnetic field given that the reading on the ammeter is $184\ \mu\text{A}$.

[3]

.....

.....

.....

.....



(c) State the direction of the current in the resistor and how you obtained this direction. [2]

.....

.....

.....

.....

(d) A student suggests that the opposing force due to the magnetic field on the moving conductor is negligible compared with other resistive forces. Is the student correct? Justify your answer with a calculation. [4]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

